

CS4423 - Networks

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0. Introduction

Lecture 1: Introduction to Networks

Connections.

Modern societies are in many ways **highly connected**. Certain aspects of this phenomenon are frequently described as **networks**. This course is meant as an introduction into the emerging interdisciplinary subject of **network science**.

Lectures

- This term, **all lectures will be online**, at the scheduled times:
- Wednesday at 9am and Thursday at 2pm
- Some lecture material will be pre-recorded videos.
- Lecture slides and links to videos can be found on the CS4423 blackboard page.

Assignments

- There will be some homework assignments, approximately 4 over the semester.
- The assignments will be marked and contribute a **continuous assessment** mark worth 40% of the overall score.
- The remaining 60% come from the **Summer exam**.
- The exam will have 4 questions, each worth 15%.

Useful Books.

- Vito Latora, Vincenzo Nicosia and Giovanni Russo.
Complex Networks.
Cambridge UP 2017.
- Mark Newman.
Networks: an Introduction.
Oxford UP 2018.
- Albert-L'aszl'o Barab'asi.
Network Science.
Cambridge UP 2016.
- Ernesto Estrada and Philip A. Knight.
A first course in network theory.
Oxford UP 2015.
- David Easley and Jon Kleinberg.
Networks, Crowds and Markets.
Cambridge UP 2010.

Examples of Networks?

...

Examples of Networks.

Newman broadly divides the most commonly studied real world networks into four classes:

- **technological** networks,
- **social** networks,
- **information** networks and
- **biological** networks.

There is of course some **overlap** between these classes. But it is an interesting exercise to list some examples within each class, and to describe their **general structure**.

- In each case, a network connects parts of a system (**nodes**) by some means (**links**).
- Different techniques used to **display**, **discover** and **measure** the structure in each example.

Technological Networks.

- Technological networks rely on a **physical infrastructure**.
- In many cases, this infrastructure has been built over many decades
- It forms part of the **backbone** of modern societies.
- This includes
 - **road** and other **transportation networks**,
 - **power grids**,
 - the **telephone network**, and more recently
 - the **Internet**.

Power Grid.



Transmission System
400 kV, 275 kV, 220 kV and 110 kV
October 2007

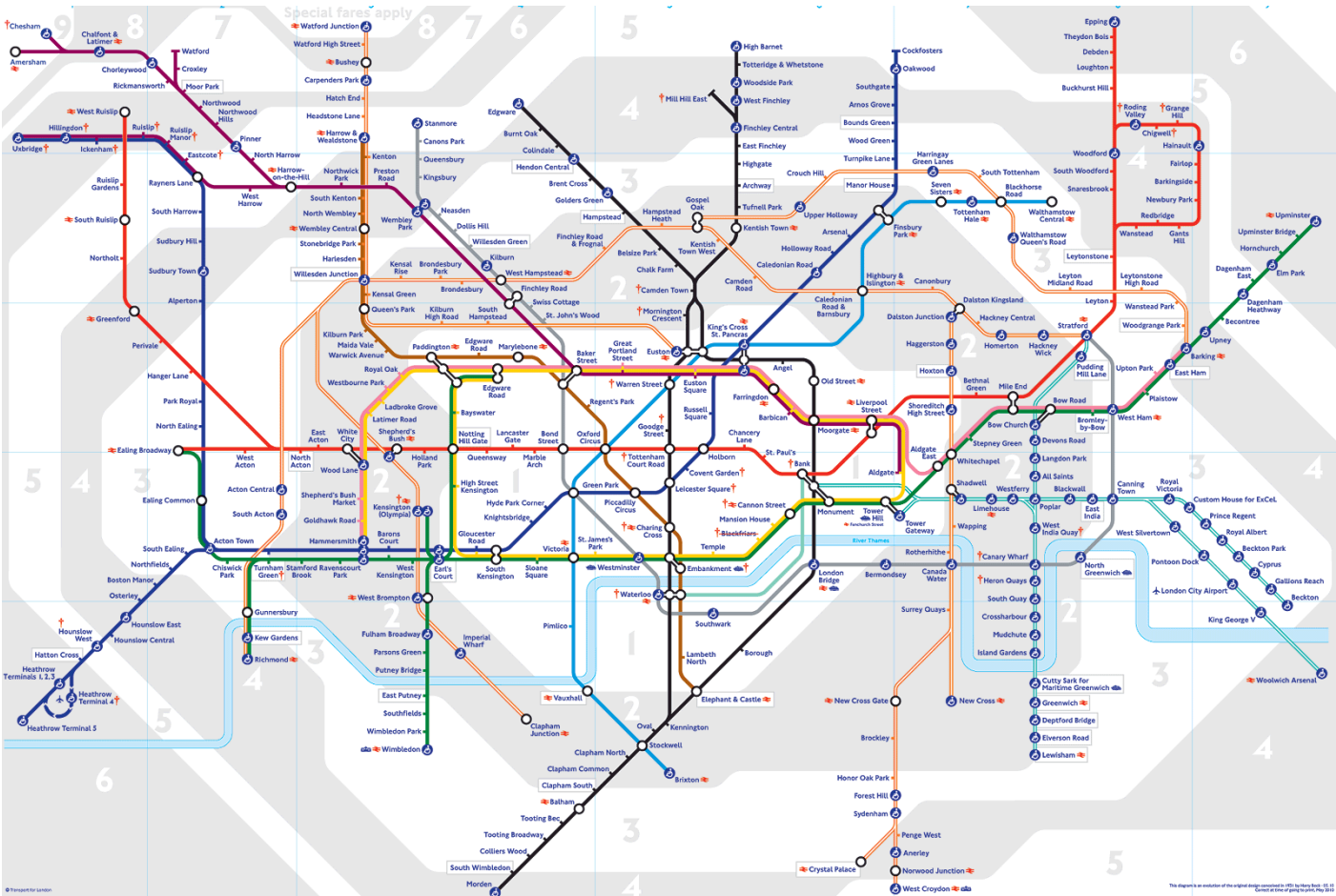
- 400 kV Lines
- 275 kV Lines
- 220 kV Lines
- 110 kV Lines
- 330 kV Cables
- 110 kV Cables
- 400 kV Stations
- 275 kV Stations
- 220 kV Stations
- 110 kV Stations
- ⚡ Phase Shifting Transformer

Transmission
Connected
Generation

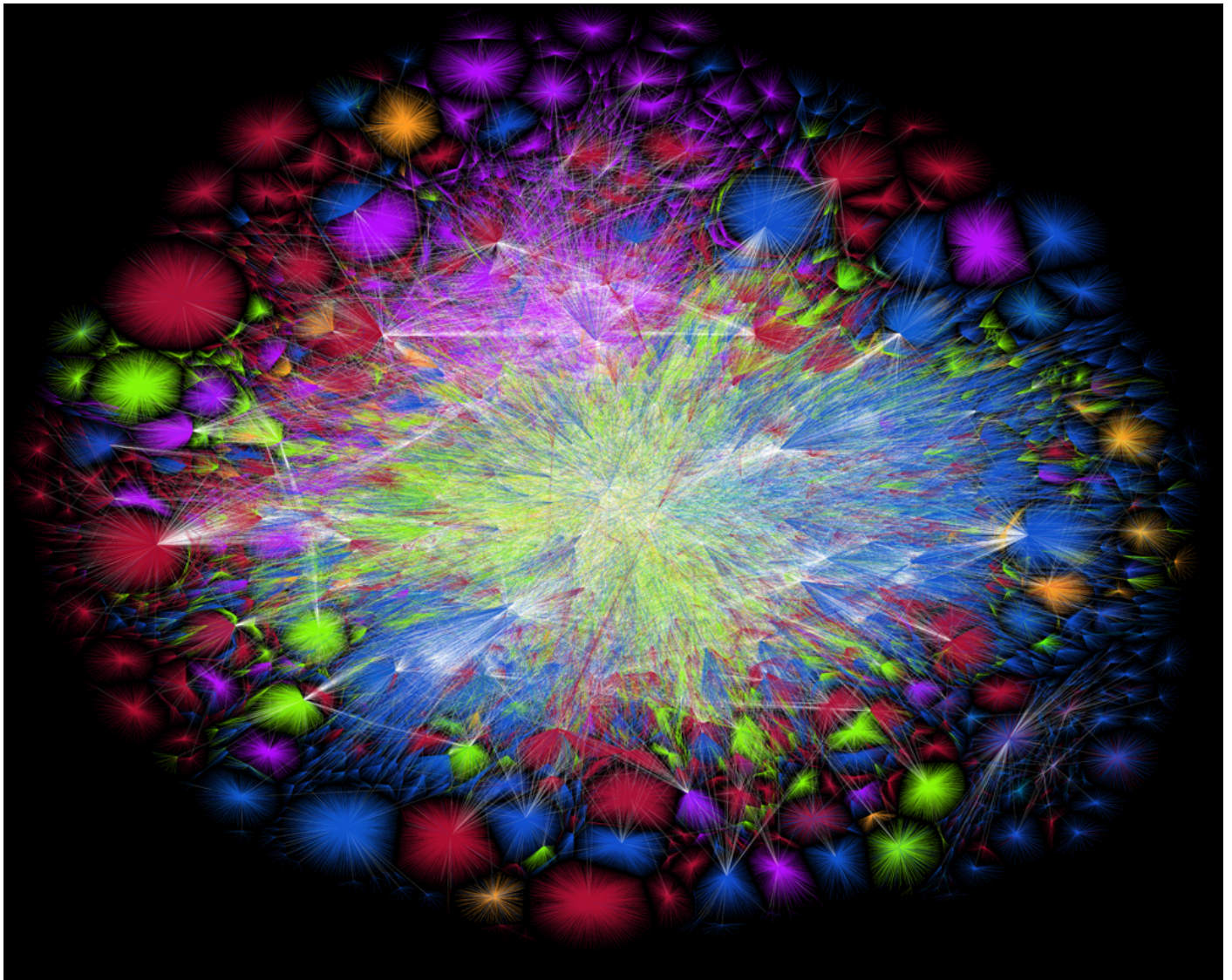
- Hydro Generation
- Thermal Generation
- ▼ Pumped Storage Generation
- Wind Generation



Transportation networks.



The Internet 2015.



(www.opte.org)

Social Networks.

- The vertices of a social network are **people**, with edges representing some sort of **social interaction**.
- In **sociology**, the vertices are often called **actors**, and the edges are called **ties**.
- A social network's **existence** does not depend on social networking sites like **facebook** or **twitter**.
- Sociologists have studied social networks long before people started exhibiting their relations to others online.
- Traditionally, data about the structure of social networks have been compiled by **interviewing** the people involved.

Information Networks.

- An **information network** consists of **data items** which are linked to each other in some way.
- Example: **relational databases**.
- Sets of information (like **scientific publications**) have been linking to each other (e.g., through **citations**) long before computers were invented.
- However, links in digital form are easier to follow ...

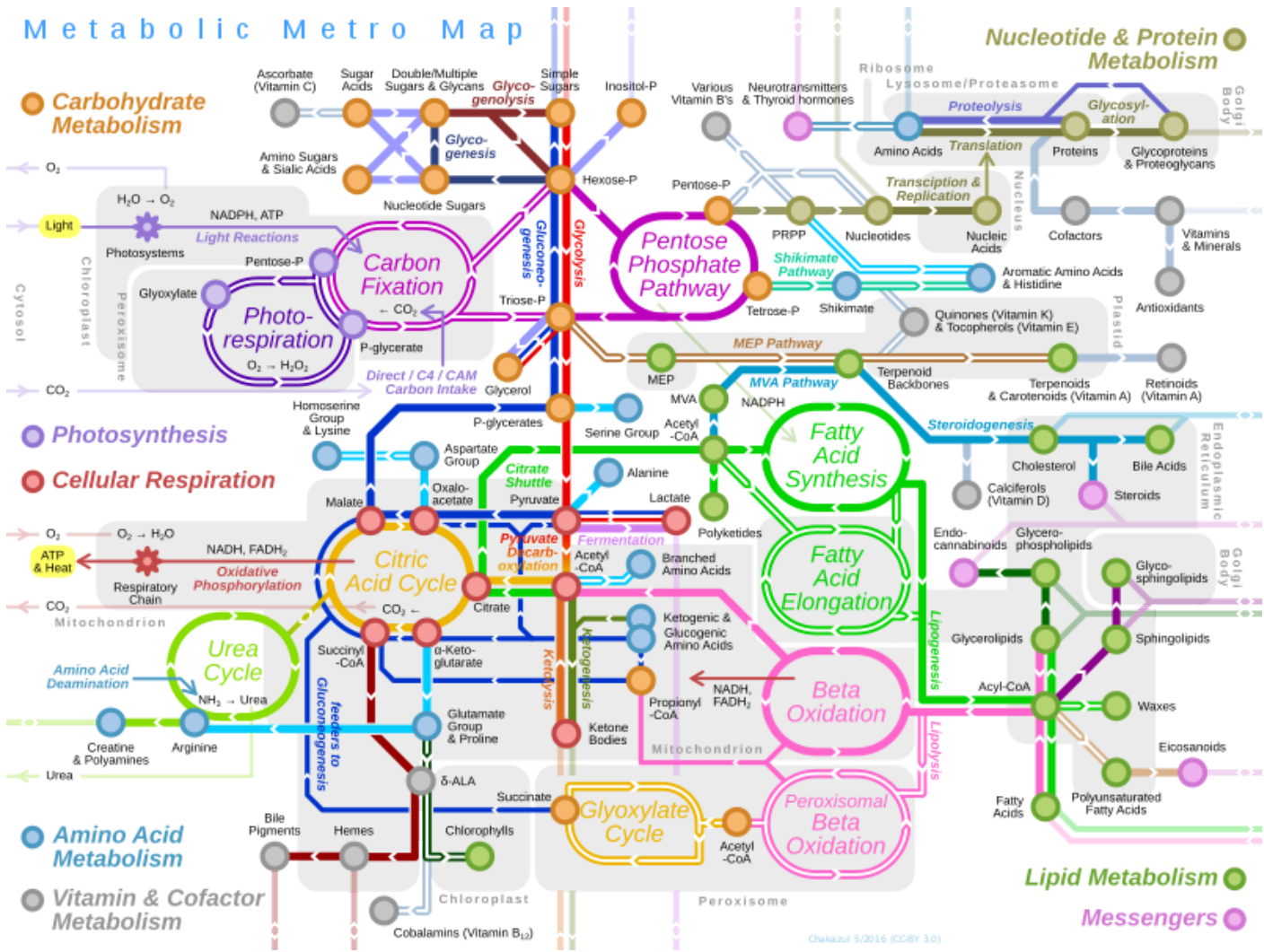
The (World Wide) Web.

- The **WWW** is probably the most wide spread and best known examples of an information network.
- Its nodes are **web pages** containing information in form of text and pictures, and its edges are the **hyperlinks**. allowing us to surf, or **navigate** from page to page.
- Hyperlinks run in **one direction only**, from the page that contains the hyperlink to the page that is referenced.
- Therefore, the WWW is a **directed network**, a graph where each edge has a direction.

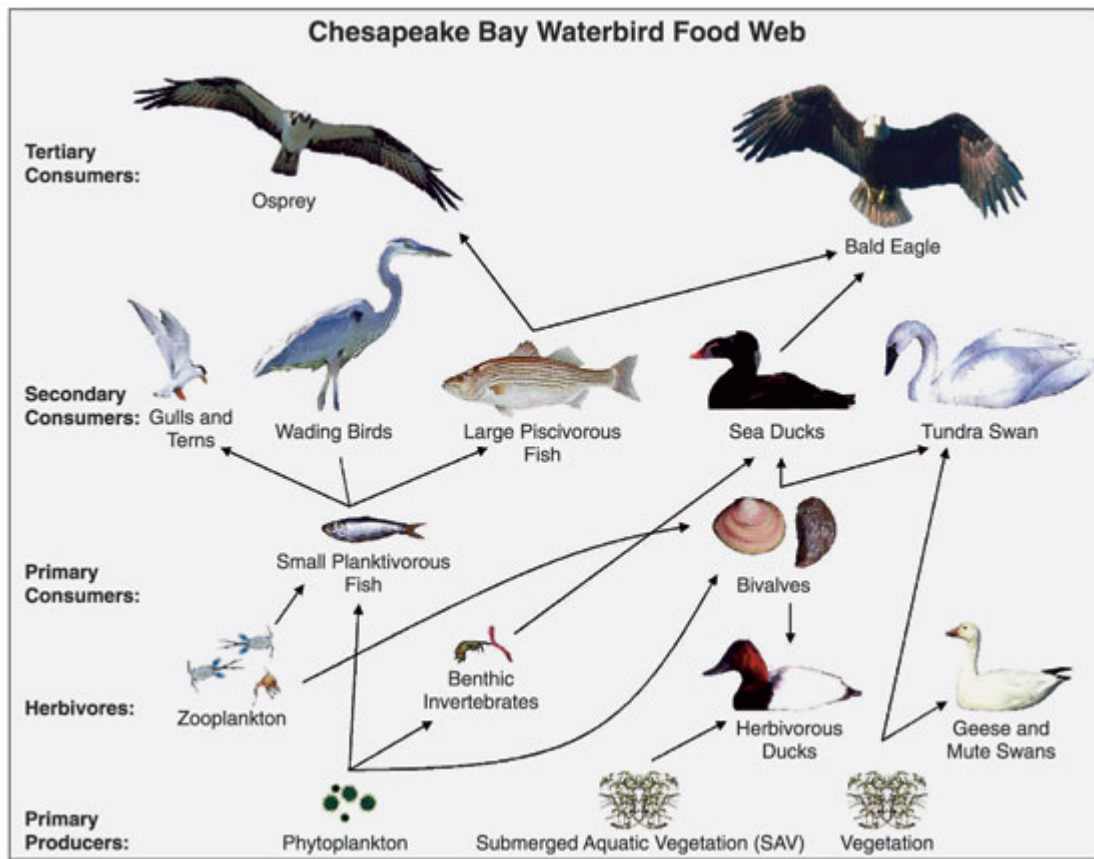
Biological Networks.

- **Biochemical networks** represent molecular level patterns of interaction and control mechanisms in the biological cell:
 - **metabolic** networks,
 - **protein-protein interaction** networks and
 - **genetic regulatory** networks.
- A **neural network** can be represented as a set of vertices, the neurons, connected by two types of directed edges, one for excitatory inputs and one for inhibiting inputs.
- **Ecological networks** are networks of ecological interactions between species.

Metabolic Networks



Food Web



Simple Forms of Networks

- In its simplest form, a network is just a collection of points (called **vertices** or **nodes**), some of which are joined in pairs (called **edges** or **links**).
- Many systems of interest are composed of individual parts that are in some way linked together.
- Such systems can be regarded as networks, and thinking about them in this way can often lead to new and useful **insights**.
- **Network science** studies the **pattern of connections** between the components of a system.
- Naturally, the **structure** of the network can have a big impact on the **behavior** of a system.
- The connections in a **social network**, for example, affect how people learn, form opinions, or spread a disease ...

Network Science

- A network is a **simplified representation** of a complex system by vertices and edges.
- The scientific study of networks is an **interdisciplinary** undertaking that combines ideas from
 - mathematics,
 - computer science,
 - physics,
 - the social sciences,
 - biology
 - ...
- Between these scientific fields, many **tools** have been developed for **analyzing, modeling** and **understanding** networks.

Network Measures

- An example of a useful and important type of network measures is that of **centrality**.
- It is concerned with the question of how **important** a particular vertex or edge is in a networked system.
- Different concepts have suggested for capturing **mathematically** what it means to be central.
- For example, a simple measure of the centrality of a vertex is its **degree**, i.e., the number of edges it is part of.
- Which measurements and calculations give **meaningful answers** for a particular system depends of course on the specific nature of the system and the questions one wants to ask.

Network Concepts

- Another interesting network concept is the **small-world effect**.
- It is concerned with the question of how far apart two randomly chosen points in a network typically are.
- Here, **distance** is usually measured by the number of edges one would need to cross over, when travelling along a **path** from one vertex to the other.
- In real world social networks the distance between people tends to be rather small.
- This observation is known as the **six degrees of separation** in popular culture.

Contents

This course is meant as an **introduction to Network Science** where, within the given time and space constraints, some but certainly not all of its interesting aspects will be discussed.

1. Graphs and Graph Theory
2. Centrality Measures
3. Random Graphs
4. Small Worlds
5. Power Laws and Scale-Free Graphs
6. Models of Growing Graphs

Jupyter Notebooks and other Software

- Lecture notes and Assignments come in the form of `jupyter` notebooks.
- This allows us to include interactive `python` code with the text.
- The notebooks will be uploaded (and updated) on `github` at

<https://github.com/gpfeiffer/cs4423> (<https://github.com/gpfeiffer/cs4423>)

- If `jupyter` is installed on your computer, you can download the notebooks and execute them there.
- Alternatively, you can follow the links



(<https://mybinder.org/v2/gh/gpfeiffer/cs4423/master>) or



(<https://colab.research.google.com/github/gpfeiffer/cs4423>) to execute the notebooks on `binder`, or Google's `colab` site ...

Networkx

We will use the `networkx` package and other packages from the `python` scientific stack to work on nontrivial examples of networks.

```
In [1]: import networkx as nx  
        nx.__version__
```

```
Out[1]: '2.5'
```

`networkx` allows us to quickly construct graphs and to work with their elements.

```
In [2]: G = nx.Graph(["AB", "BC", "BD", "CD"])  
        list(G.nodes)
```

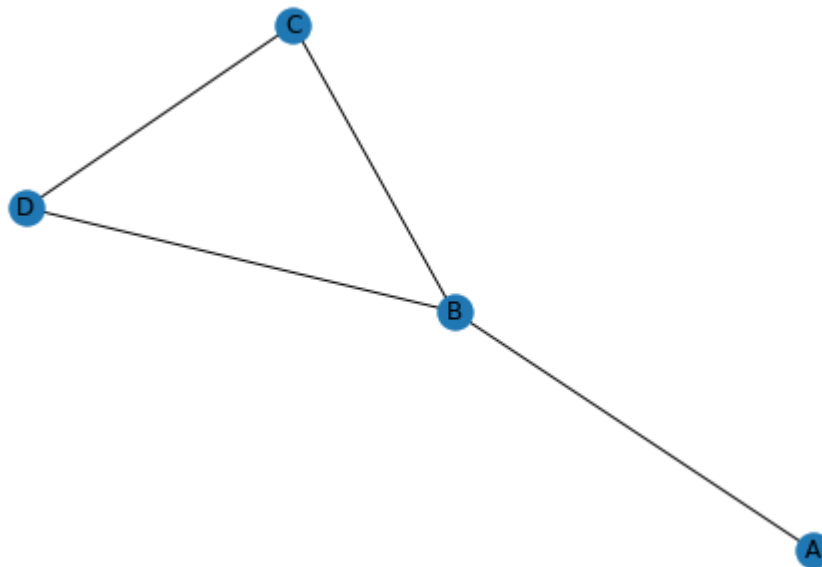
```
Out[2]: ['A', 'B', 'C', 'D']
```

```
In [3]: list(G.edges)
```

```
Out[3]: [('A', 'B'), ('B', 'C'), ('B', 'D'), ('C', 'D')]
```

The `matplotlib` package contains the functionality needed to draw a diagram of the graph on this page.

```
In [4]: nx.draw(G, with_labels=True)
```



More about graphs and their implementation in the next lecture ...

Summary

We hope to develop a **network perspective** as a powerful way of looking at **complex systems** in general. Networks provide a way of thinking about

- social dynamics,
- economic interaction,
- online information,
- designed technology and
- natural processes,

and approaching such systems with an eye toward their **internal structural patterns** and the resulting **rich feedback effects**.

Code Corner

python

- The `import` statement [\[doc\] \(https://docs.python.org/3/reference/simple_stmts.html#import\)](https://docs.python.org/3/reference/simple_stmts.html#import) provides access to packages, or their parts and elements.

```
In [5]: import math  
math.pi
```

```
Out[5]: 3.141592653589793
```

```
In [6]: from math import pi  
pi
```

```
Out[6]: 3.141592653589793
```

```
In [7]: import numpy as np  
np.array([1,2,3])
```

```
Out[7]: array([1, 2, 3])
```

- The `list` constructor [\[doc\] \(https://docs.python.org/3/library/stdtypes.html#list\)](https://docs.python.org/3/library/stdtypes.html#list) tries and creates a list from its arguments.

```
In [8]: list("networks")
```

```
Out[8]: ['n', 'e', 't', 'w', 'o', 'r', 'k', 's']
```



```
In [9]: list((1,2,3,4))
```

```
Out[9]: [1, 2, 3, 4]
```

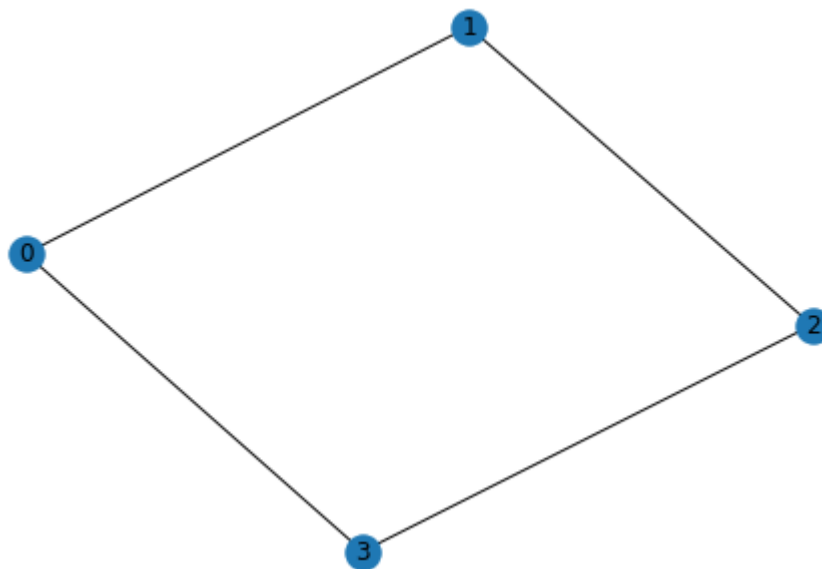
networkx

- The Graph constructor [\[doc\]](https://networkx.github.io/documentation/networkx-1.10/tutorial/tutorial.html) (<https://networkx.github.io/documentation/networkx-1.10/tutorial/tutorial.html>) can be used to construct a graph from data in various formats.

```
In [10]: edges = ["01", "12", "23", "30"]  
G = nx.Graph(edges)
```

- The draw function [\[doc\]](https://networkx.github.io/documentation/networkx-1.10/reference/drawing.html) (<https://networkx.github.io/documentation/networkx-1.10/reference/drawing.html>) depends on the `matplotlib` library.

```
In [11]: nx.draw(G, with_labels=True)
```



Exercises

- Import the `sin` function from the `math` package and use it to compute $\sin(\pi/4)$.
- Construct and draw a graph with 5 nodes and 5 edges that form a circle.

```
In [ ]:
```